01: Intro Stats Review

Stat 230 - Spring 2026

Researchers are interested in whether how much time young adults spend sitting each day (reading, watching TV, using a computer, commuting) is related to symptoms of depression. To study this, they used data from the National Health and Nutrition Examination Survey (NHANES), an ongoing survey run by the CDC that collects health and behavioral data from a nationally representative, randomly selected sample of the U.S. population.

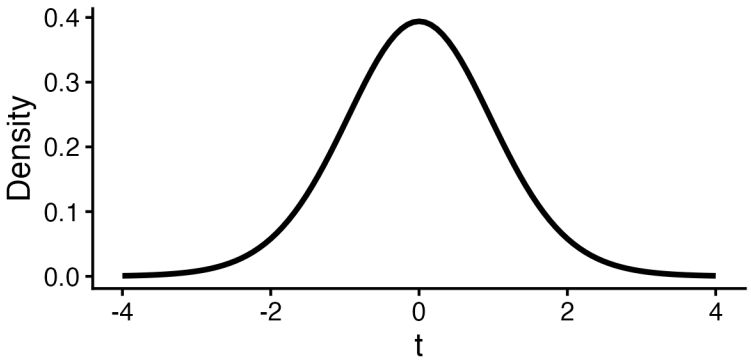
Using data from the NHANES survey, participants aged 18-25 were classified into two groups: those who reported **more** sitting time than average ("High"), and those who reported **less** ("Low"). Their depression symptoms were measured with the PHQ-9, a standard 9-item screening scale (0-27, higher = more symptoms).

sitting_group	n	xbar	sd
High	300	3.58	4.36
Low	359	3.21	4.04

1. What is the null hypothesis for this test?

2. What is the alternative hypothesis for this test?

3. The researchers conduct a two-sample t-test and report a test statistic of $t = 1.26$. Below is a plot of the appropriate t-distribution for this test. Sketch how you would calculate the p-value for this situation.



4. How did you calculate this p-value in your last statistics course?

5. If the p-value you calculated was 0.26, what would you conclude about the null hypothesis?

6. Was this an observational study or an experiment? Based on your answer, can you conclude that spending more time sitting *causes* higher depression symptoms? Why or why not?

7. Can you generalize these results to all young adults in the U.S.? Why or why not?